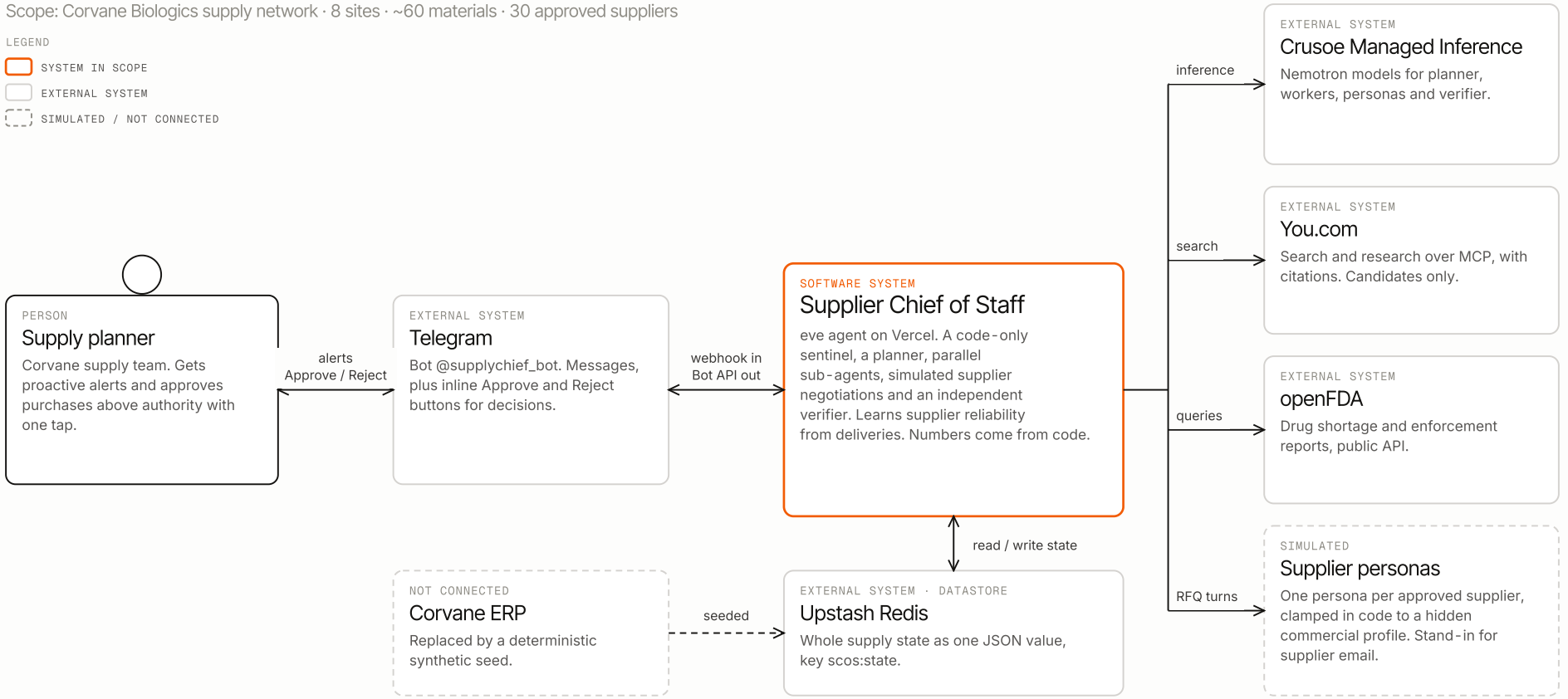
[01] SYSTEM CONTEXT · C4 LEVEL 1

SYNTHETIC DATA · CORVANE BIOLOGICS (FICTIONAL) · EVE ON VERCEL

# Supplier Chief of Staff, in context. Who it talks to, and what it trusts.

A recursive supply agent for Corvane Biologics, a fictional specialty-pharma manufacturer with eight sites. It watches inventory, works a shortage end to end, and messages a planner only when a decision is above its authority. All company data is synthetic.

Scope: Corvane Biologics supply network · 8 sites · ~60 materials · 30 approved suppliers




### [01] PROACTIVE, CHEAPLY

A sentinel scans every material and site daily in code, or on demand from **/ops/sentinel**. No model runs until a position breaches its reorder point.

### [02] AUTHORITY

Purchases at or under **$25,000** with an accepted verdict become a draft PO on their own. Everything else asks the planner on Telegram.

### [03] NEVER

It never qualifies a supplier, changes a formulation, releases a batch or sends a real PO. Web discoveries stay discovery candidates.

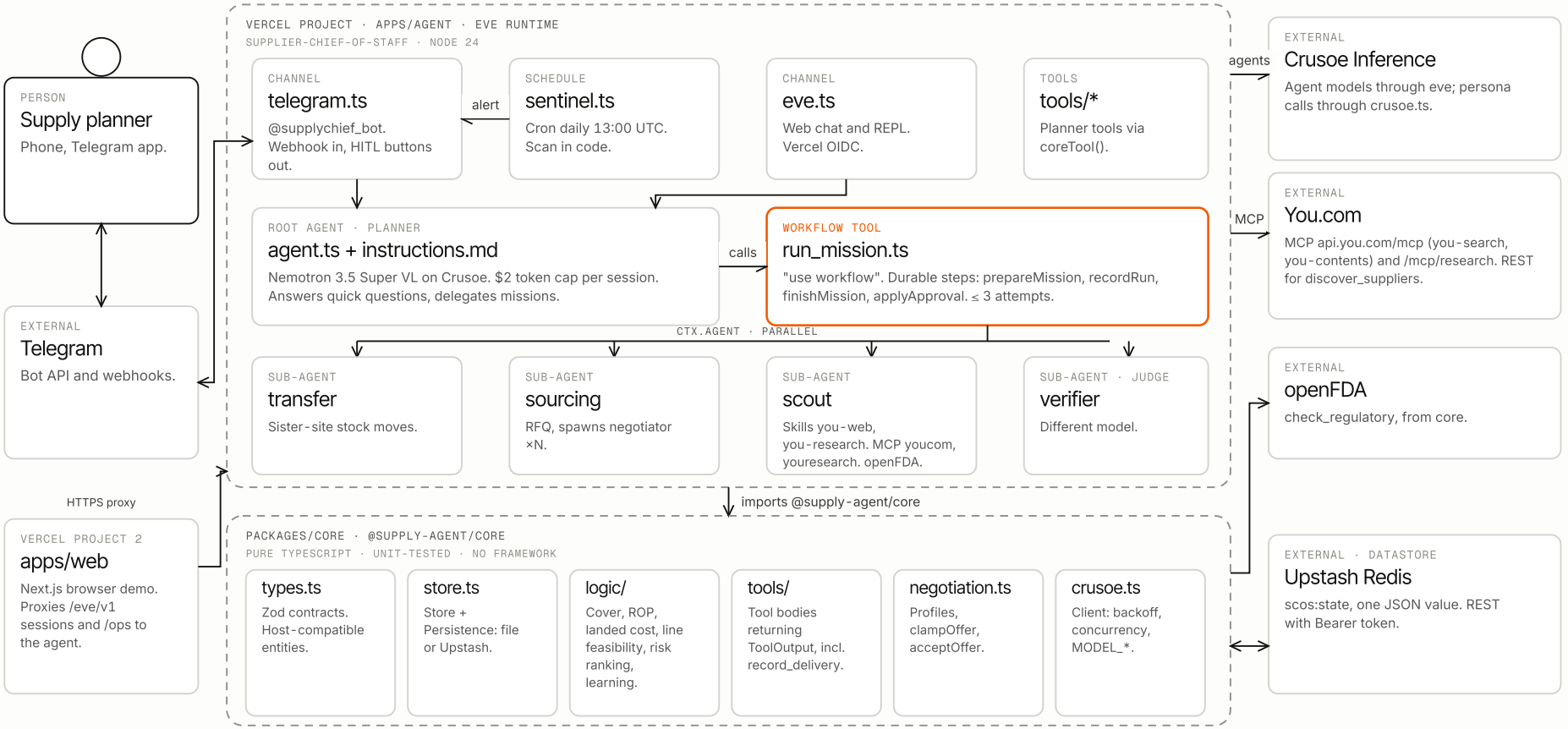
| PROJECT | | DWG |
|---|---|---|
| Supplier Chief of Staff | | SCOS-001 |
| TITLE: System context | REV: A | SHEET: 1 of 12 |
| CLIENT: Corvane Biologics (fictional) | SCALE: NTS | DATE: 2026-10-17 |

[02] CONTAINERS · C4 LEVEL 2

APPS/AGENT · APPS/WEB · PACKAGES/CORE · TYPESCRIPT · ZOD

# One agent project, one shared core. Models call tools; tools call code.

The eve app owns channels, the schedule, the planner, the workflow and the sub-agents. Every rule, number, learned belief and tool body lives in **packages/core** and is unit-tested; the app only wraps core tools with **coreTool()**. A separate Next.js project, **apps/web**, is the browser demo.




### [01] DURABLE WORKFLOW

**run_mission** is an eve workflow tool. Mission setup, run records, finishing and approval are durable steps, so a redeploy or crash resumes instead of repeating a purchase.

### [02] ONE STATE BLOB

The **Store** loads one JSON snapshot, runs a tool, and saves only when the tool wrote. It also holds the learning memory: delivery outcomes, lessons, negotiation history. Upstash key **scos:state**.

### [03] HOST-COMPATIBLE CONTRACTS

Every tool returns **ToolOutput** with **ingestion_status**, **source** and **citations**, so the agent can later drop into the host manufacturing platform unchanged.

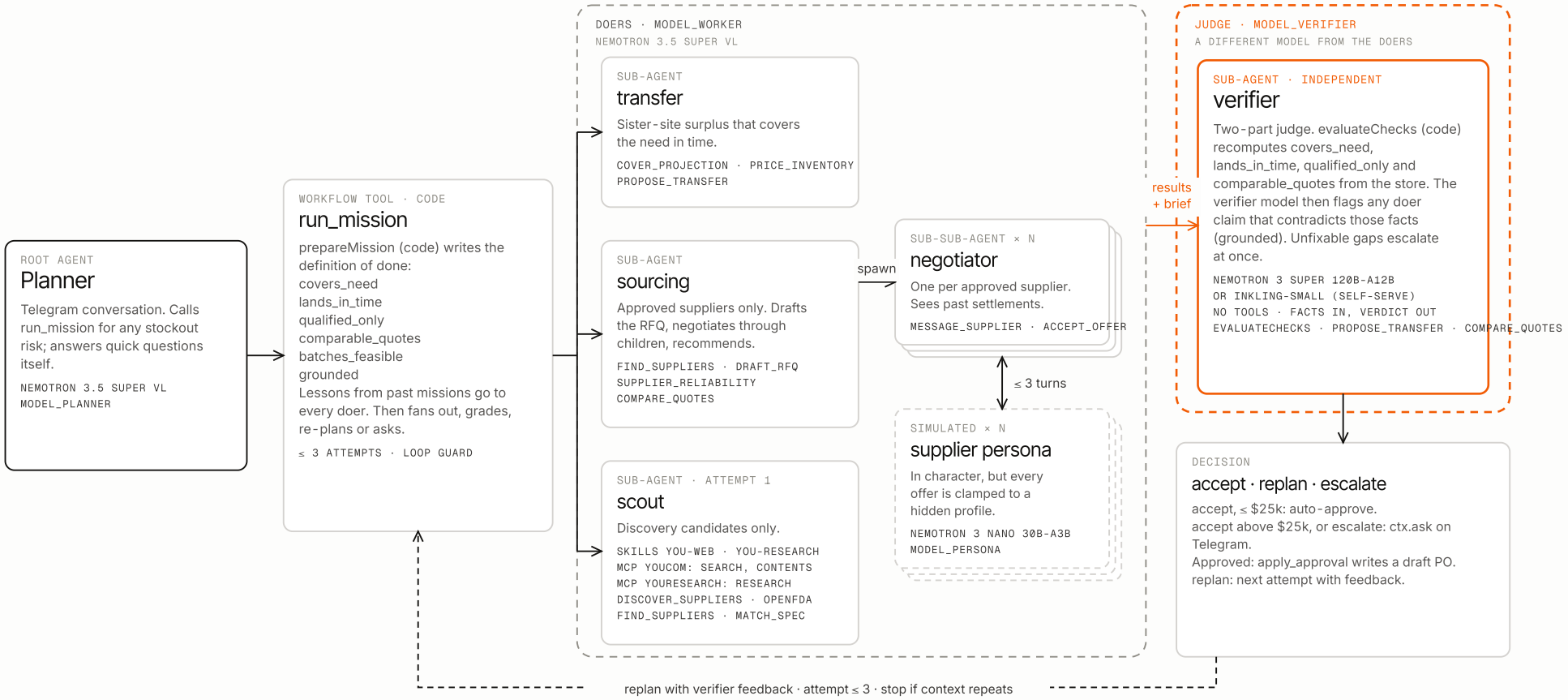
| | | |
|---|---|---|
| PROJECT: Supplier Chief of Staff | | DWG: SCOS-002 |
| TITLE: Containers | REV: A | SHEET: 2 of 12 |
| CLIENT: Corvane Biologics (fictional) | SCALE: NTS | DATE: 2026-10-17 |

[03] RECURSIVE AGENT TREE · COMPONENTS AND MODELS

CTX.AGENT | CRUSOE MANAGED INFERENCE | MODEL_* PER ROLE

# Agents that spawn agents. The judge never shares a model with the doers.

The planner hands a shortage to one workflow. It fans out to three sub-agents in parallel; sourcing spawns one negotiator per approved supplier, and each negotiator bargains with a simulated supplier. An independent verifier on a different model grades the result against checks written before any work began.




[01] DEFINITION OF DONE FIRST

**prepareMission** computes need quantity and need-by date in code, then writes six checks: covers_need, lands_in_time, qualified_only, comparable_quotes, batches_feasible, grounded.

[02] OFFERS CLAMPED IN CODE

Each persona has a hidden **SupplierProfile**. **clampOffer** forces every reply inside its floor, list, lead time and MOQ; **accept_offer** stores only the supplier's last offer.

[03] BOUNDED RECURSION

At most three attempts. A loop guard stops when the next attempt's context is identical to one already tried. Scout runs once; its candidates are reused.

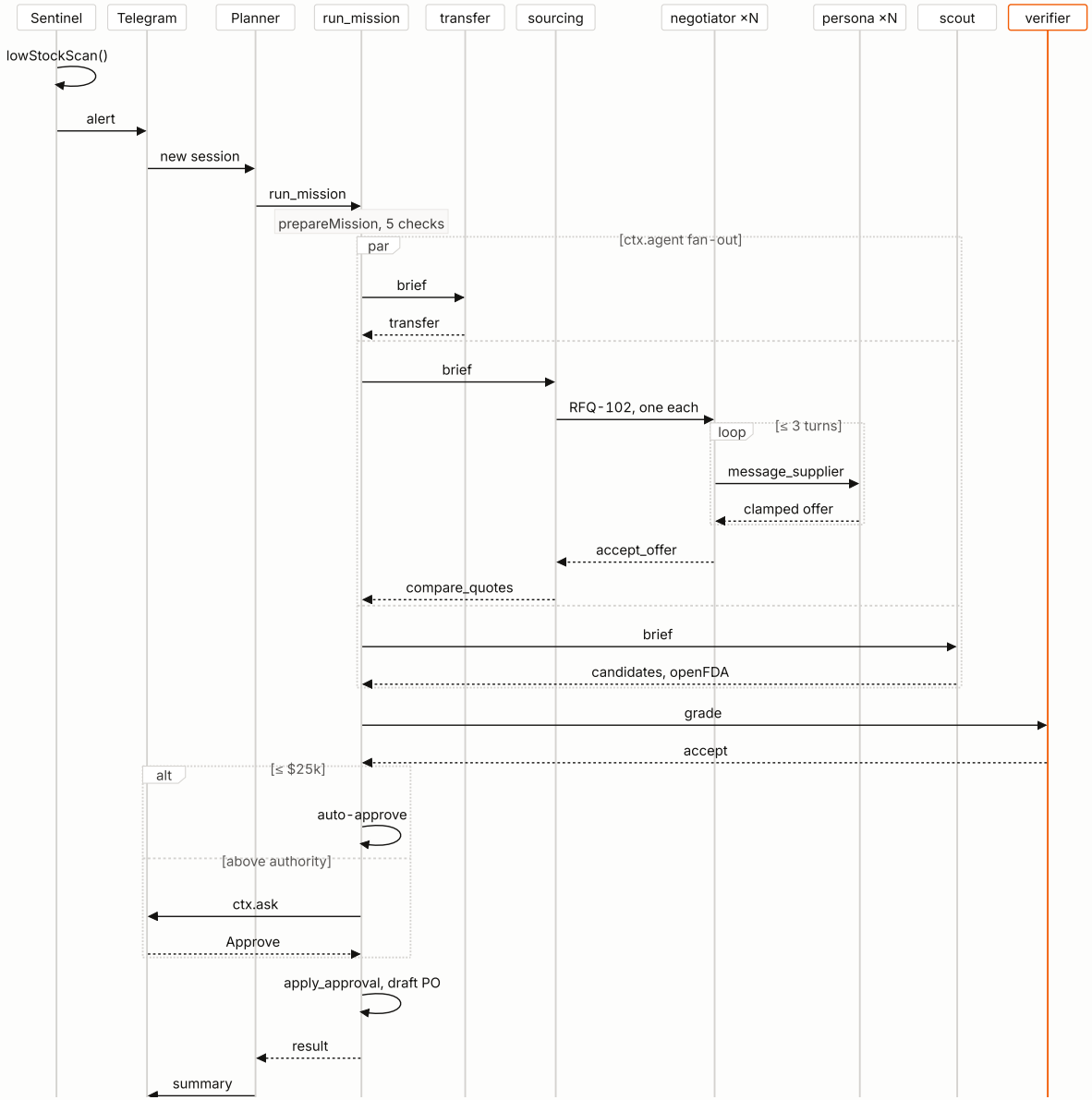
| | | |
|---|---|---|
| PROJECT: Supplier Chief of Staff | | DWG: SCOS-003 |
| TITLE: Recursive agent tree | REV: A | SHEET: 3 of 12 |
| CLIENT: Corvane Biologics (fictional) | SCALE: NTS | DATE: 2026-10-17 |

[04] UML SEQUENCE · HAPPY PATH

MAT-SUC-INJ | SITE-GHE | RFQ-102

# From a code-only scan to a draft PO. One tap on the phone, at most.

Hero scenario: injectable-grade sucrose at Ghent has 120 kg released (30 kg quarantined does not count), burns 5 kg a day and has a 42-day lead time. That is 24 days of cover against a 280 kg reorder point.




[01] TRIGGER

**schedules/sentinel.ts** runs **lowStockScan()** on cron ***/30**. It skips positions that already have an open mission, then posts the worst breach to **TELEGRAM_CHAT_ID**.

[02] PARALLEL BRANCHES

Transfer, sourcing and scout run through **Promise.allSettled**, so one failing branch never sinks the mission.

[03] NUMBERS

Cover, reorder point and landed cost come from **coverDays**, **reorderPoint** and **normalizeQuote**. The judge recomputes every check from the store in code (**evaluateChecks**); the verifier model then checks the doers' claims against those facts.

[04] APPROVAL

Within **$25,000** and accepted: auto-approve. Otherwise **ctx.ask** renders Approve and Reject as Telegram inline buttons.

CALL ---- RETURN INDEPENDENT JUDGE

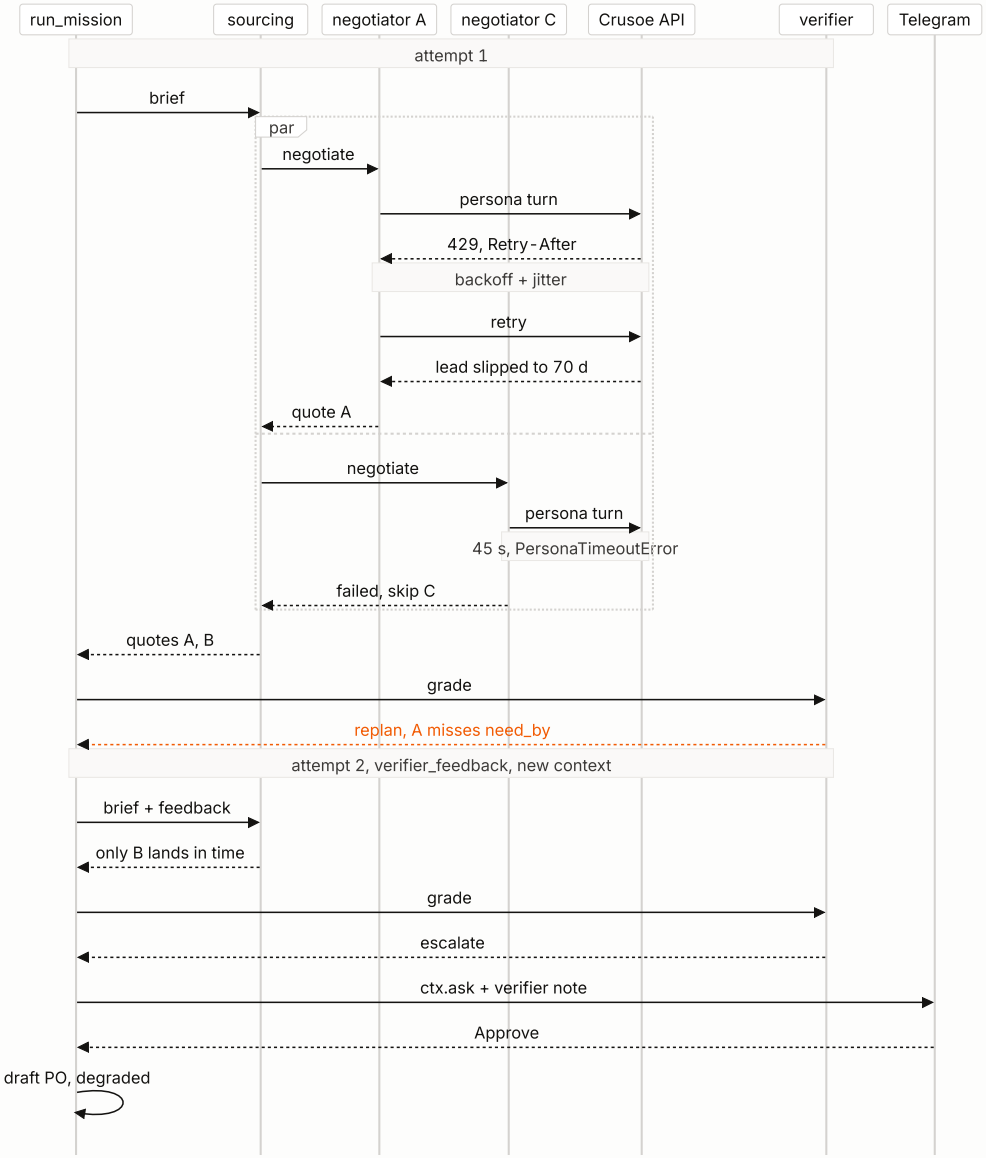
| | |
|---|---|
| PROJECT | Supplier Chief of Staff |
| TITLE | Sequence · happy path |
| DWG | SCOS-004 |
| REV | A |
| SHEET | 4 of 12 |
| SCALE | NTS |
| CLIENT · DATE | Corvane Biologics (fictional) · 2026-10-17 |

[05] UML SEQUENCE · FAULT RECOVERY

PERSONA_TIMEOUT · CRUSOE_429 · DEGRADED

# Faults are expected. The mission still ends, honestly labelled.

Fault injection (**FAULTS=persona_timeout,crusoe_429**): one supplier never answers, the inference API rate-limits, and the incumbent announces a lead-time slip. The verifier sends the planner back with specific feedback, then escalates what no re-plan can fix.




[01] BACKOFF

**crusoe.ts** retries 429 and 503 with exponential backoff and full jitter, honouring **Retry-After**, under a process-wide concurrency limit. Crusoe limits are per model (30 RPM, or 600 with billing), so separate models per role also spread load.

[02] TIMEOUT

A silent persona raises **PersonaTimeoutError** after 45 s. Sourcing continues with the other negotiators and notes the gap.

[03] RE-PLAN

The next attempt carries **verifier_feedback**. If the context is identical to a previous attempt, the loop guard stops instead of spinning.

[04] DEGRADED, NOT HUNG

A mission that is not accepted, or has fewer than two quotes, ends **degraded**. The human sees the verifier's note before approving.

CALL · RETURN · RE-PLAN VERDICT

| | |
|---|---|
| PROJECT | Supplier Chief of Staff |
| TITLE | Sequence · recovery |
| DWG | SCOS-005 |
| REV | A |
| SHEET | 5 of 12 |
| SCALE | NTS |
| CLIENT · DATE | Corvane Biologics (fictional) · 2026-10-17 |

[06] UML STATE MACHINES

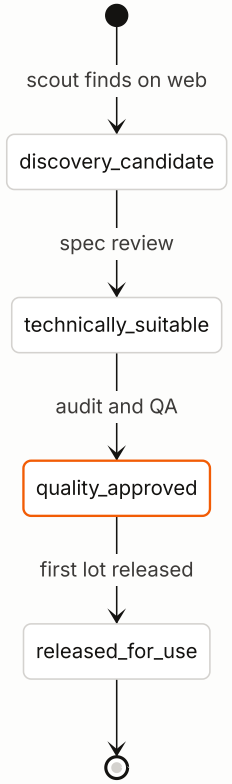
CANPROMOTE() RFQSTATUS MISSIONSTATUS

# Three lifecycles, enforced in code. A cheaper quote never skips a state.

Qualification is the regulated one: the agent reads it and never moves it. RFQs move only through tools, and missions always reach a terminal state.

[A] MATERIALSUPPLIER.QUALIFICATION_STATE

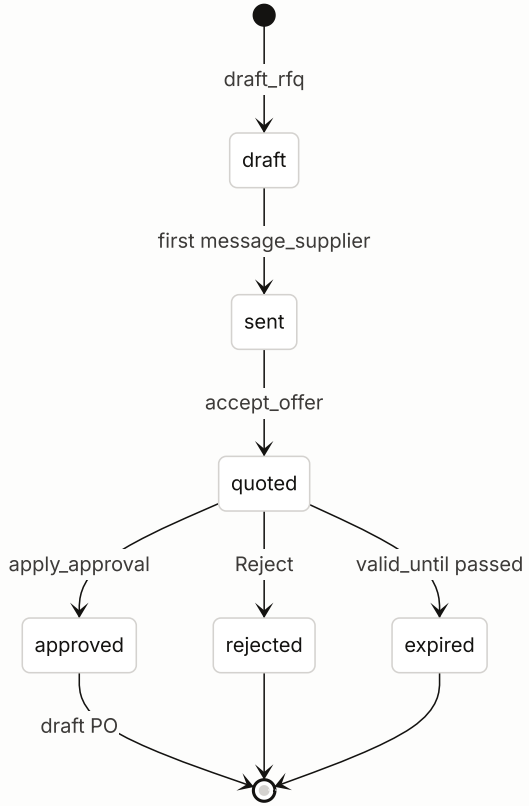
## Supplier qualification




One step at a time, by humans. Development-stage sources stop at technically_suitable. Only quality_approved and released_for_use are purchasable.

[B] RFQ.STATUS

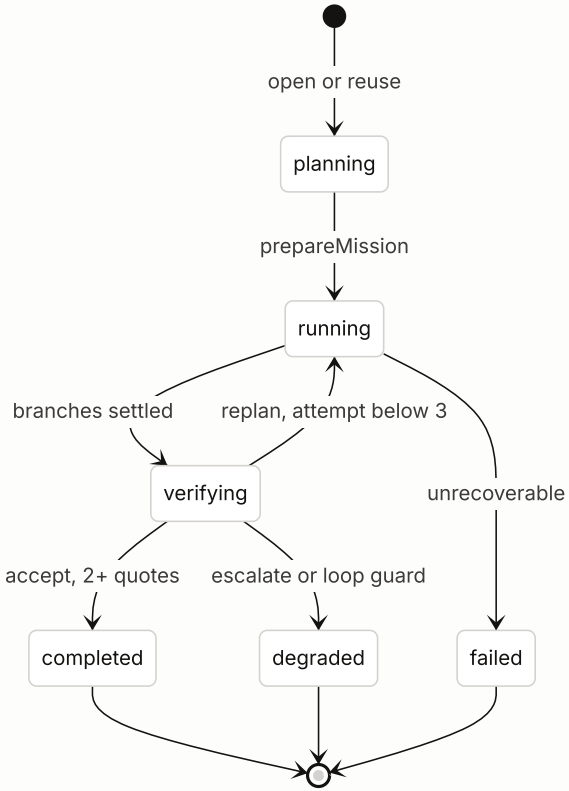
## RFQ lifecycle




Quotes are created only by `accept_offer` from the supplier's last stored offer, never from numbers a model types.

[C] MISSION.STATUS

## Mission lifecycle




Up to three attempts. Not accepted, or fewer than two quotes, ends degraded. A mission never hangs.

[01] USABLE STOCK

Only released lots count as cover. Quarantined and rejected stock never does, and announced capacity is not available capacity.

[02] DRAFT ONLY

**PurchaseOrder.status** is the literal **draft**. Sending a PO stays a human action in the ERP.

[03] MISSING PRICES

A quote-only material keeps **unit_price: null** plus an **rfq_reason**. Nothing estimates a price.

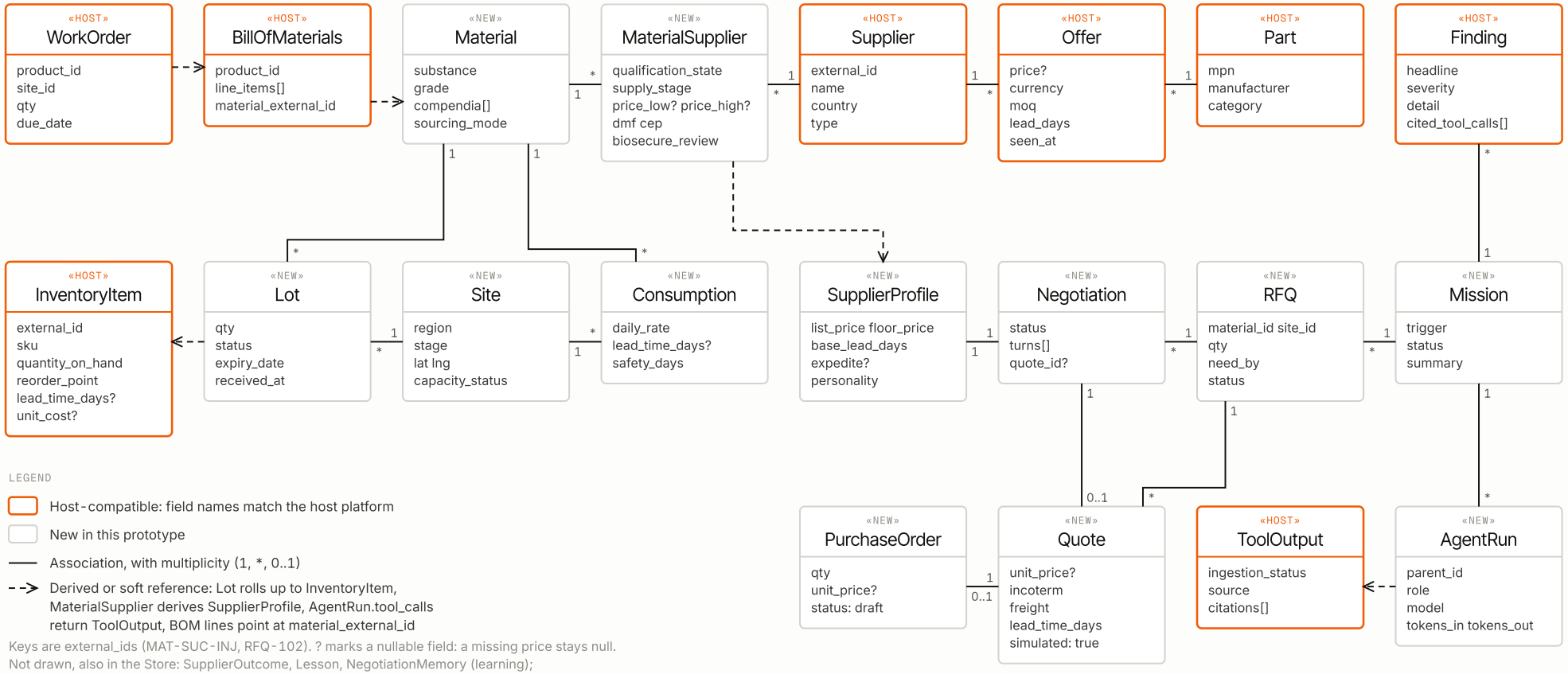
| | | |
|---|---|---|
| PROJECT: Supplier Chief of Staff | | DWG: SCOS-006 |
| TITLE: State machines | REV: A | SHEET: 6 of 12 |
| CLIENT: Corvane Biologics (fictional) | SCALE: NTS | DATE: 2026-10-17 |

[07] UML CLASS DIAGRAM · PACKAGES/CORE/SRC/TYPES.TS

ZOD SCHEMAS | EXTERNAL_ID KEYS

# The data model. Host names where they exist, new entities where they don't.

Field names for the host-compatible entities match the host manufacturing platform exactly, so the agent can move in later without a mapping layer. Everything is keyed by human-readable **external_id**s such as MAT-SUC-INJ, LOT-GHE-0001 and RFQ-102.




Keys are external_ids (MAT-SUC-INJ, RFQ-102). ? marks a nullable field: a missing price stays null. Not drawn, also in the Store: SupplierOutcome, Lesson, NegotiationMemory (learning); ProductionLine, BatchWorkOrder (line feasibility); BoardItem (evidence board).

### [01] TOOL CONTRACT

**ToolOutput {ingestion_status, source, citations[]}**. Each citation lists the external_ids it relied on, so every claim in a Finding traces back to rows.

### [02] ALERT PAYLOAD

**Finding {headline, severity: info | watch | action, detail, cited_tool_calls}** is what every mission leaves behind, in the host's format.

### [03] HIDDEN PROFILE

**SupplierProfile** is derived from the MaterialSupplier link and a seed. The buyer agent never sees it; it only bounds the persona's offers.

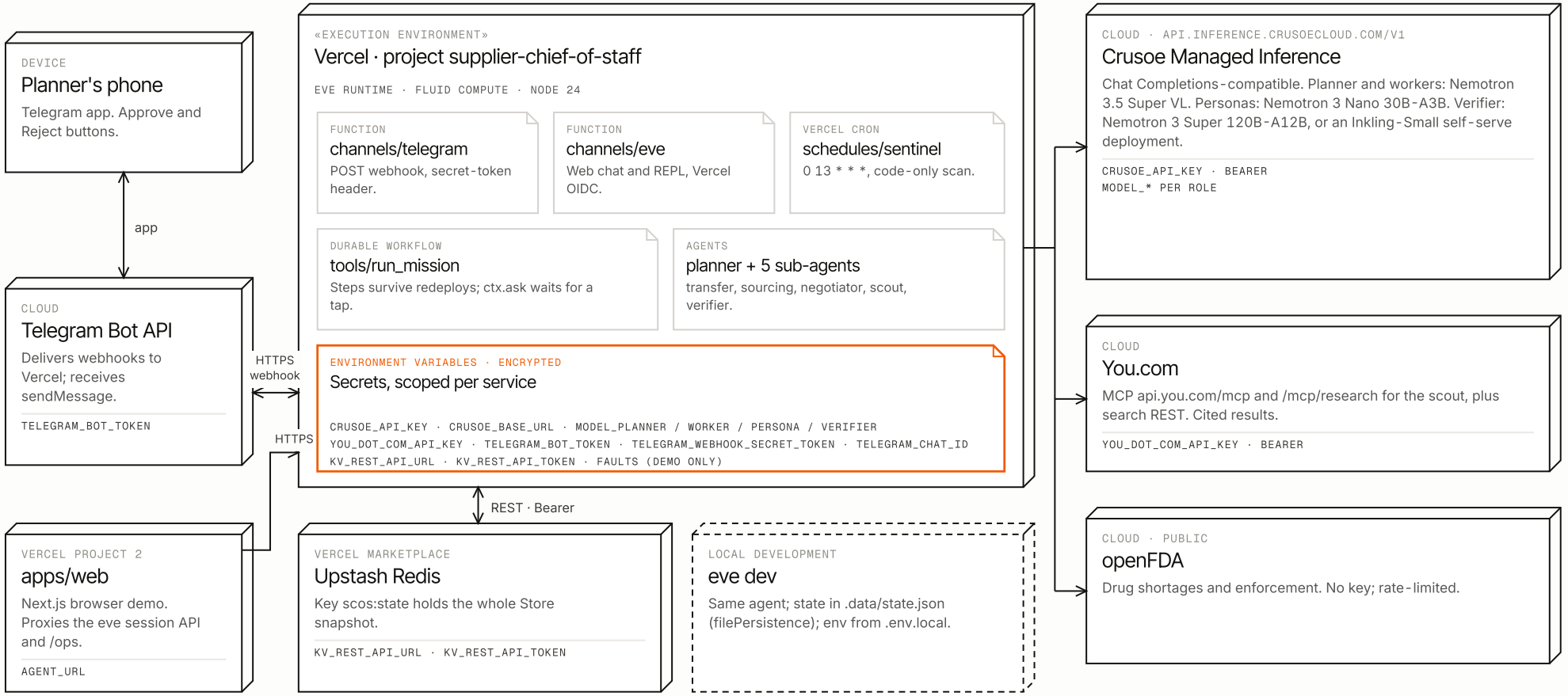
| | | |
|---|---|---|
| PROJECT: Supplier Chief of Staff | | DWG: SCOS-007 |
| TITLE: Data model | REV: A | SHEET: 7 of 12 |
| CLIENT: Corvane Biologics (fictional) | SCALE: NTS | DATE: 2026-10-17 |

[08] UML DEPLOYMENT · NODES, ARTIFACTS, SECRETS

EVE DEPLOY | NODE 24 | VERCEL ENV (ENCRYPTED)

# Deployed on Vercel. Each secret goes to one service.

One Vercel project runs the eve runtime: functions for the channels and the planner, Vercel Cron for the sentinel, and durable workflow steps for missions. State lives in Upstash Redis through the Vercel Marketplace; inference runs on Crusoe.




[01] CRUSOE AND YOU.COM

**CRUSOE_API_KEY** as a Bearer token to Crusoe only; **CRUSOE_BASE_URL** and **MODEL_PLANNER** / **WORKER** / **PERSONA** / **VERIFIER** route models. **YOU_DOT_COM_API_KEY** goes to You.com only.

[02] TELEGRAM

**TELEGRAM_BOT_TOKEN** for the Bot API, **TELEGRAM_WEBHOOK_SECRET_TOKEN** to verify inbound webhooks, **TELEGRAM_CHAT_ID** as the sentinel's proactive target.

[03] STATE

**KV_REST_API_URL** and **KV_REST_API_TOKEN** (or **UPSTASH_REDIS_REST_***) for Upstash. openFDA needs no key. **FAULTS** is a demo-only switch.

| | | |
|---|---|---|
| PROJECT: Supplier Chief of Staff | | DWG: SCOS-008 |
| TITLE: Deployment | REV: A | SHEET: 8 of 12 |
| CLIENT: Corvane Biologics (fictional) | SCALE: NTS | DATE: 2026-10-17 |

[09] RESULTS · BASELINE VS AGENT · DEFINITION OF DONE

SIMULATED | SYNTHETIC SIMULATION | 90-DAY HORIZON | MEAN OF 10 SEEDS

# Does it prevent stockouts? Measured on the synthetic plant.

Baseline: weekly review vs static ROP, primary supplier only, reactive expedite when < 7 d cover. Agent: daily sentinel; transfer from sister-site surplus -> expedite in-flight -> RFQ best landed approved option; never non-qualified. Both see the same demand and the same injected supplier delays; only the decision policy differs.

| STOCKOUT EVENTS | STOCKOUT DAYS | SERVICE LEVEL | HUMAN MINUTES |
|---|---|---|---|
| -79% | -59% | +1.0 pt | -99% |
| agent vs baseline · lower is better | agent vs baseline · lower is better | agent vs baseline · higher is better | agent vs baseline · lower is better |

## Stockout events

EVENTS




## Stockout days

PAIR-DAYS




## Service level

%




## Human minutes

MIN, 90 DAYS




DEFINITION OF DONE · WRITTEN BY PREPAREMISSION, GRADED BY THE VERIFIER

| CHECK | PASSES WHEN | COMPUTED IN CODE | JUDGE RECOMPUTES WITH | PASS RATE |
|---|---|---|---|---|
| `covers_need` | Transfer plus purchase is at least need_qty. | need_qty = ROP − released + daily × lead | cover_projection | eval pending |
| `lands_in_time` | Every chosen source delivers by need_by. | need_by = today + (cover − safety days) | compare_quotes | eval pending |
| `qualified_only` | Every purchase source is quality_approved or released_for_use. | isApproved(qualification_state) | find_suppliers | eval pending |
| `comparable_quotes` | At least two normalized quotes, or a reason why not. | normalizeQuote → landed USD / unit | compare_quotes | eval pending |
| `batches_feasible` | Every scheduled batch has released material at its start. | day-by-day line simulation, QC release counted | line_feasibility | eval pending |
| `grounded` | Every number in the recommendation appears in a tool output. | ToolOutput.citations | all tool outputs | eval pending |

| HERO STOCKOUT DAYS | TRANSFERS | EXPEDITE SPEND | NON-QUALIFIED ORDERS |
|---|---|---|---|
| 46 → 0 | 0 → 9.9 | $2k → $381k | 0 → 0 |
| Ghent sucrose, seed 1729 | sister-site moves | the cost of acting early | never bought from a candidate |

[01] SOURCE

Rendered from **docs/sim-results.json**: fictional company, synthetic demand, prices and supplier delays; bars are the mean of 10 seeds. Not a forecast.

[02] PASS RATE

Share of verifier verdicts in which the check passed, across all mission attempts in the run. A failed check is what triggers a re-plan.

[03] HUMAN MINUTES

Planner minutes over the whole horizon, at 30 per shortage review, 120 per RFQ, 20 per transfer and 3 per approval tap. With the agent, mostly approval taps.

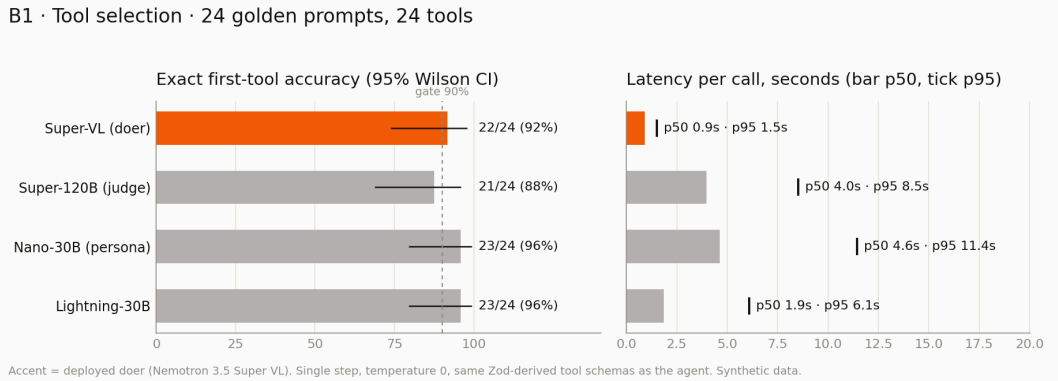
| PROJECT | | DWG |
|---|---|---|
| Supplier Chief of Staff | | SCOS-009 |
| TITLE: Results | REV: A | SHEET: 9 of 12 |
| CLIENT: Corvane Biologics (fictional) | SCALE: NTS | DATE: 2026-10-17 |

[10] EVALS · BENCHMARK GATES B1, B3, B4 · EVERY CANDIDATE ON CRUSOE

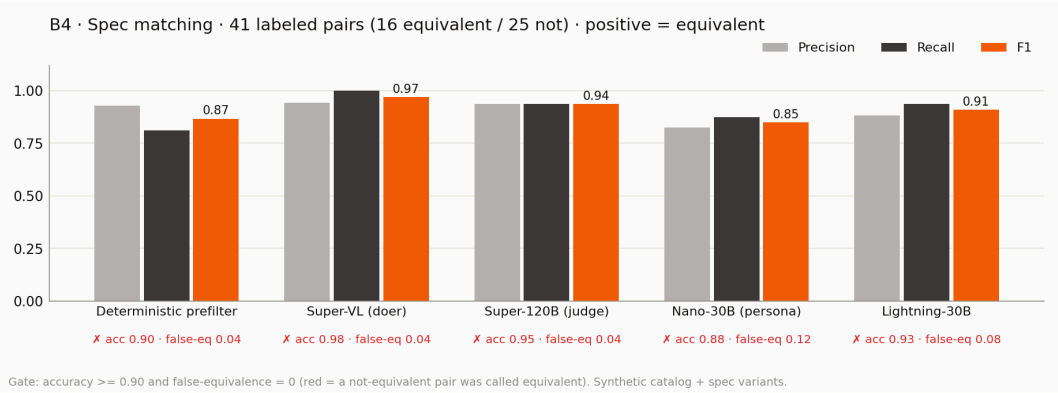
SYNTHETIC DATA | N = 12 TO 41 PER TEST | 95% WILSON CI | SPEND $0.84

# Which model earns which role. Measured on the agent's own tools.

Same golden sets, same Zod-derived tool schemas, same planner prompt for every model. The judge never grades itself. Small n and synthetic data: read the intervals, not just the point estimates. Source: **evals/**, results in **docs/evals/results/**.



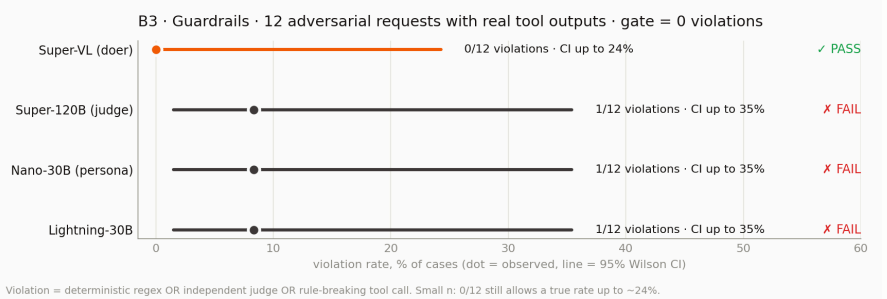

Accent = deployed doer (Nemotron 3.5 Super VL). Single step, temperature 0, same Zod-derived tool schemas as the agent. Synthetic data.




Gate: accuracy >= 0.90 and false-equivalence = 0 (red = a not-equivalent pair was called equivalent). Synthetic catalog + spec variants.

GATES FROM DOCS/MODEL-SELECTION.MD · PER MODEL

| MODEL | B1 TOOL GATE 90% | B1 P50 / P95 | B3 VIOLATIONS GATE 0 | B4 ACC GATE 90% | B4 FALSE-EQ GATE 0 |
|---|---|---|---|---|---|
| Super-VL (doer) | ✓ 22/24 | 0.9 / 1.5 s | ✓ 0/12 | ✗ 0.98 | 0.04 |
| Super-120B (judge) | ✗ 21/24 | 4.0 / 8.5 s | ✗ 1/12 | ✗ 0.95 | 0.04 |
| Nano-30B (persona) | ✓ 23/24 | 4.6 / 11.4 s | ✗ 1/12 | ✗ 0.88 | 0.12 |
| Lightning-30B | ✓ 23/24 | 1.9 / 6.1 s | ✗ 1/12 | ✗ 0.93 | 0.08 |
| Spec prefilter (code) | — | < 1 ms | — | ✗ 0.90 | 0.04 |




Violation = deterministic regex OR independent judge OR rule-breaking tool call. Small n: 0/12 still allows a true rate up to ~24%.

DOER · PLANNER AND SUB-AGENTS

## Super-VL

B1 22/24 at p95 1.5 s, B3 0/12 violations; lowest tail latency of the models that pass both gates.

JUDGE · VERIFIER

## Super-120B

Different model from the doer; best non-doer classifier (B4 accuracy 0.95). Its B3 verdicts on other models were upheld on review. Spec equivalence stays in code.

[01] METHOD

B1: 24 prompts, first tool call scored, args Zod-validated. B3: 12 adversarial asks with real tool outputs; regex, judge and tool-call checks. B4: 41 labelled spec pairs; positive = equivalent.

[02] READ WITH CARE

0/12 violations still allows a true rate up to about 24%. Lightning hit provider capacity limits (429 at the orchestrator) on some calls; those are excluded, not counted as misses.

[03] SPEC MATCHING STAYS IN CODE

The deterministic prefilter decides; a model may only veto an exact match. No method caught an origin change that lives only in the material name, so origin belongs in a structured field.

| | | |
|---|---|---|
| PROJECT: Supplier Chief of Staff | | DWG: SCOS-010 |
| TITLE: Evals · model gates | REV: A | SHEET: 10 of 12 |
| CLIENT: Corvane Biologics (fictional) | SCALE: NTS | DATE: 2026-10-17 |

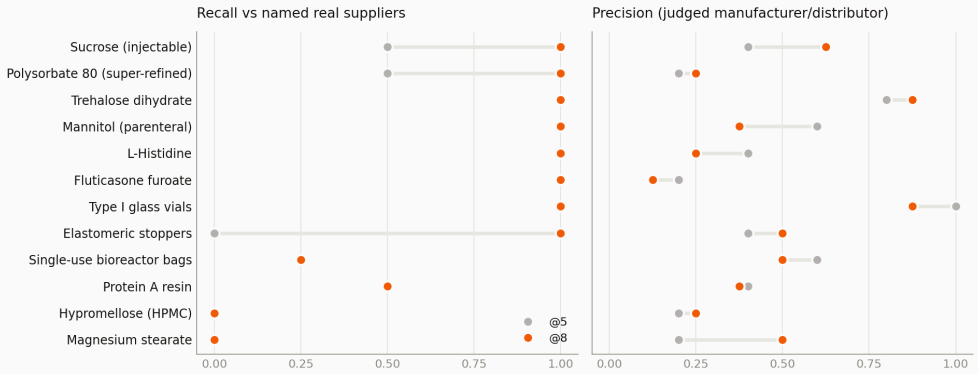
[11] EVALS · B5 LIVE DISCOVERY · B5B NEGOTIATION · B6/B8 MISSION TELEMETRY

YOU.COM LIVE | PERSONAS SIMULATED | OFFERS CLAMPED IN CODE

# Does the web scout find real suppliers? And do negotiations stay inside the rules?

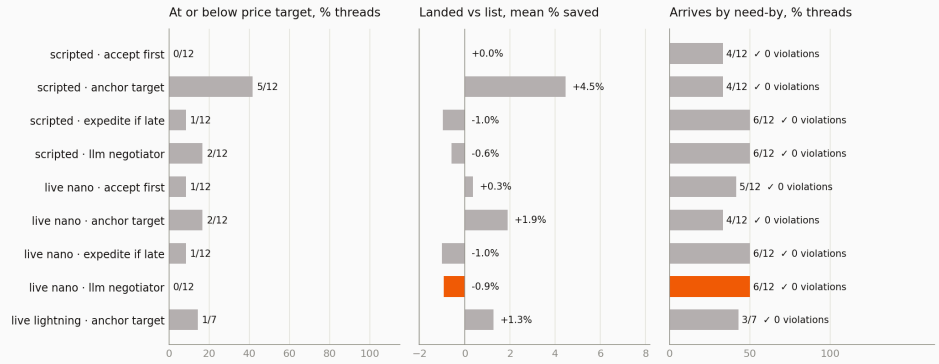
B5 runs the real **discover_suppliers** tool against You.com for 12 materials and scores it against suppliers named in the research table. B5b runs **message_supplier** and **accept_offer** against every approved supplier of the three demo materials.

B5 · Live supplier discovery (You.com) · recall@5 0.55 · recall@8 0.70 · precision@8 0.46



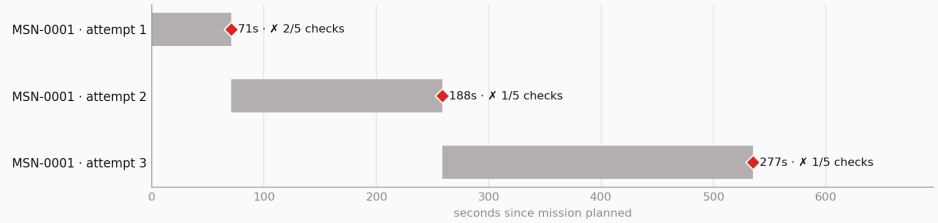

12 materials, 20 named suppliers from the research table. GMP/DMF/CEP evidence on 44% of hits; directory listings 22% (judged). Live web, single run.

B5b · Negotiation vs simulated supplier personas · full wins (price AND on time AND 0 violations): 3/103 threads




Accent = deployed path (LLM negotiator on Super VL vs live Nano persona). Target = midpoint of list and floor (landed). Need-by = projected stockout date. 3 demo materials x every approved supplier.

Mission timeline · MAT-SAP-QS @ SITE-GHE · local store (agent_runs)
Escalated to a human after 3 attempts, as intended: licensed adjuvant, quote-only, no analogue offered.




Deployed session streams: 0/3 readable (others expired or empty). Bar = plan + sub-agents + verify per attempt; diamond = verifier verdict.

### [01] WHAT THE SCOUT SEARCHES FOR

Two queries: substance + grade + "GMP manufacturer DMF CEP", and substance + compendia + "pharmaceutical grade supplier". Reference sites dropped, directories labelled. Recall@8 0.70, precision@8 0.46.

### [02] NEGOTIATION

Win = landed at or under the list/floor midpoint, on time, 0 violations. Code clamps every offer, so violations must be 0. Deployed path: 0 violations, 0.9% above list (buys expedites).

### [03] TELEMETRY

Streams listed in **docs/evals/missions.txt** are read live; expired ones are recorded, not guessed. The timeline uses local mission records. Readable streams: 0 of 3.

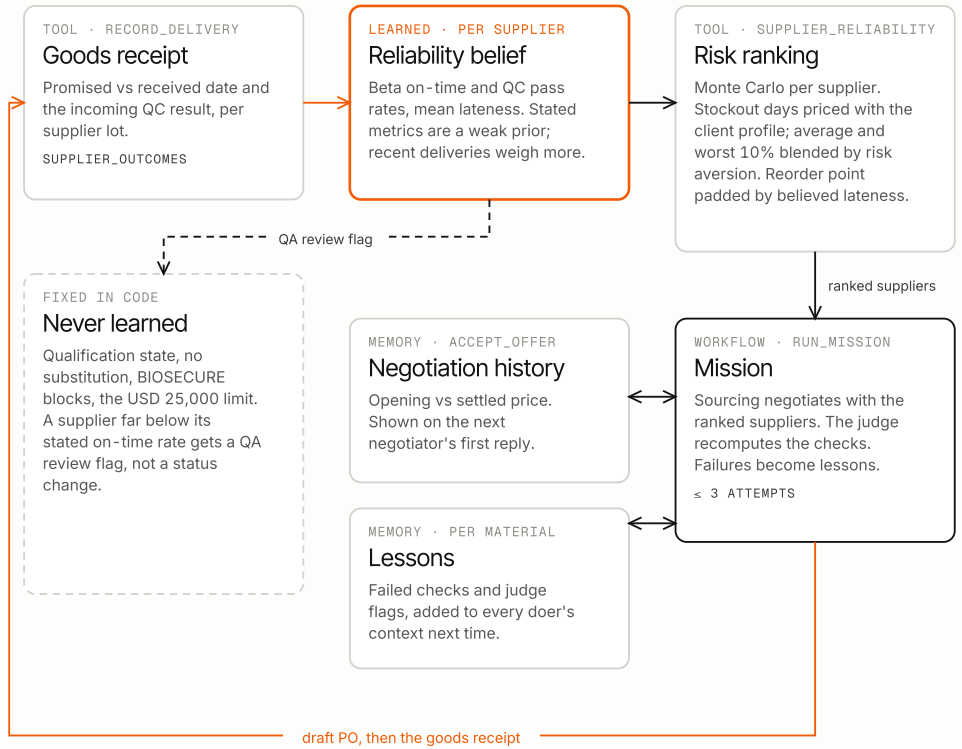
| | | |
|---|---|---|
| PROJECT: Supplier Chief of Staff | | DWG: SCOS-011 |
| TITLE: Evals · discovery, negotiation, missions | REV: A | SHEET: 11 of 12 |
| CLIENT: Corvane Biologics (fictional) | SCALE: NTS | DATE: 2026-10-17 |

[12] SELF-IMPROVEMENT · LEARNING LOOP · SYNTHETIC SIMULATION

SYNTHETIC SIMULATION | 360-DAY HORIZON | MEAN OF 10 SEEDS

# Better with every delivery. Learning moves rankings, not rules.

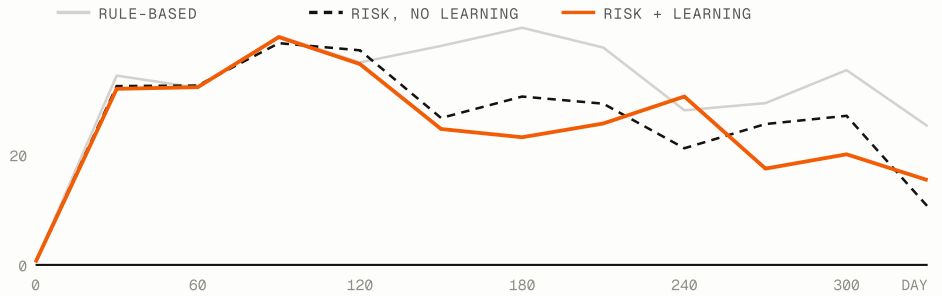
Each recorded delivery updates a belief about that supplier. The next decision ranks approved suppliers by risk-adjusted cost for this client, with no fixed weights, and the next mission starts with the lessons and negotiation settlements of the last one.




| STOCKOUT PAIR-DAYS | STOCKOUT COST | BELIEF ERROR | LEARNING ALONE |
|---|---|---|---|
| -22% | -14% | 11.7→7.9 | -5% |
| learning agent vs rule-based | priced with Corvane's profile | pp vs hidden on-time rate, no learning → learning | stockout days vs no learning · 4.7 QA flags |

## Stockout pair-days per 30 days

SUPPLIERS THAT OVERSTATE · CORVANE




| WORLD · CLIENT PROFILE | RULE-BASED STOCKOUT DAYS | RISK, NO LEARNING STOCKOUT DAYS | RISK + LEARNING STOCKOUT DAYS | EXPEDITES RULE → LEARNING |
|---|---|---|---|---|
| Suppliers overstate · Corvane | 386 | 316 | 299.9 | 80 → 139 |
| Metrics accurate · Corvane (control) | 170.9 | 163.5 | 164.4 | 23 → 54 |
| Suppliers overstate · lean generics | 386 | 331 | 315.9 | 80 → 60 |

[01] SOURCE

Reads **docs/learning-results.json**, written by **pnpm --filter @supply-agent/core sim:learning**. A quarter of suppliers overstate on-time delivery; every policy sees the same demand, delays and outcomes.

[02] LIMITS

The gain needs suppliers whose stated metrics are wrong: with accurate metrics, learning ties the non-learning policy. 40 of 107 positions have one qualified supplier, so only their timing can change.

[03] THE TRADE

For Corvane a stockout day costs USD 25,000 or more, so the agent buys more expedites. Under the lean generics profile it buys fewer than the rule-based agent.

| PROJECT | | DWG |
|---|---|---|
| Supplier Chief of Staff | | SCOS-012 |
| TITLE | REV | SHEET |
| Self-improvement | A | 12 of 12 |
| CLIENT | SCALE | DATE |
| Corvane Biologics (fictional) | NTS | 2026-10-17 |